

The Machine, Part 3: Machines che Jaal

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PART 3 CHA MINI-GLOSSARY

THE MACHINE

PART 3 OF 5: MACHINES CHE JAAL

(Ha part Marathi madhe aahe, test mhanun. Part 1 ani 2 Hinglish madhe hote. Technical shabd nehmi English madhech rahtil: server, network, address. Aavdla tar pudhche parts pan Marathi madhe hotil.)

PAACH PARTS CHA NAKASHA

PART 1	PAISA AUR MACHINE	zala
PART 2	MACHINE SE BAAT	zala
[TUMHI ITHE AAHAT]		
PART 3	MACHINES CHE JAAL	network, address, internet
PART 4	DATA AANI AATHAVNI	storage, database, shodhne
PART 5	JAGALA SERVE KARNE	scale, cloud, kharcha, purna jod

Aadhi Part 2 chi pariksha

Paach prashna, Part 2 madhle. Aadhi swataha uttar dya, mag khali baghaa. Jithe adkal, tithla chapter number sobat aahe, tithe parat ja.

1. iPhone chi app Android var ka chalat nahi? (3.5)
2. Google Android fukat ka vaatto? (3.5, 3.7)
3. Mothya software madhe bugs nakki ka astat? (4.1)
4. "Majhya machine var tar chalat hota" cha khara arth kay? (4.4)
5. Software banvtana paisa kashavar dyaycha: vachan var ki kashavar? (4.5)

Uttar: 1. App OS sathi lihileli aste, machine sathi nahi; malak (OS) vegla, mhanun app chalat nahi. 2. Vikun kamavnnya peksha MALAK hone motha khel aahe: pratyek phone Google chya search, ads ani Play Store cha darvaja hoto. 3. Situations chi sankhya bomb sarkhi vadhte; jo jod lihinaryachya dokyat aalach nahi, toch kadhi tari user la bhetto. Hi ginti aahe, halgarjipana nahi. 4. Software = code + tyacha aaspaas cha mahol (OS, libraries, settings). Code same, mahol vegla = veglach natija. 5. Chaltia tukdyavar. Vachan nahi, demo. Pratyek tukda CHALUN dakhavla pahije.

Ya part madhe kay aahe

Aata paryant sagli goshta EKA machine chya aat hoti. Pan tumcha pratyek message dusrya machine kade jato: shahara palikde, samudra palikde, ardhy second madhe. Kasa?

Aath chapters: machines ekmekanna kashya shodhtat (address), message kasa pravaas karto (packets), doghanni niyam aadhich ka tharvave lagtat (protocols), naav numbers kashe hotat (DNS), public rastyavar private goshta kashi lapte (encryption), internet mhanje nakki kay, aani tyache niyam kon banavto.

He tumchya business decision madhe kuthe yeil: Tumcha pratyek product internet var chalel. Tyacha speed, tyacha kharcha, tyachi suraksha, he sagle ithlya samajni var ubhe aahe. Ani ek motha dhanda-secret ithe lapla aahe: protocols ani networks madhe “n squared” naavachi ek problem parat parat yete, ani jo ti sodavto to darvajacha malak hoto. UPI he tyache jivant udaharan aahe, ya part madhe bhettel.

SECTION 5: MACHINES CHE AAPSAT BOLNE

Aath chapters. Eka machine pasun jagbhar pasarlelya jaalaparyant.

He tumchya business decision madhe kuthe yeil: “App slow ka aahe,” “server cha kharcha kiti yeil,” “data safe aahe ka,” “he feature offline chael ka”: he sagle prashna network chya samajni shivay sutat nahit. Ani jo founder network samajto, to he pan baghto ki darvaje (platforms) kase bantat: internet chya itihasat jo jo “sagle ekmekanshi bolu shaktil” asa rasta banvto, to to shrimant hoto. Ya section madhe to pattern char vela distil.

Chapter 5.1 [SPINE]: Ek machine puri padat nahi

Part 1 ani 2 ni tumhala ek shaktishali machine dili: universal machine, jya var kuthlihi recipe chalte. Mag prashna asa: ekach machine puri ka padat nahi? WhatsApp tumchya phone var aahe na?

Nahi. Tumchya phone var WhatsApp cha fakt DARVAJA aahe. Vichar kara: tumhi message pathavla ani mitracha phone band aahe. Tari message pohochto, to phone chalu kartach. Mhanje message madhe kuthe tari THAMBLA hota. Kuthe? Tumchya phone var nahi (tumhi pathavla, gela). Tyachya phone var nahi (band hota). Mhanje madhye TISRI machine aahe, jya var message vaat baghat basla hota.

Ti tisri machine WhatsApp chya company chi aahe, ani ashya machines lakhoni aahe. Ka lagtat? Teen karana, tinhi tumhi swataha kadhu shakta:

Data sagl-yancha ekatra asava lagto. Tumcha bank balance tumchya phone var thevla tar? Phone haravla, paisa gela. Ani dukandarala kasa kalel ki tumchyakade paise aahe? Jo data DOGHANNA lagto, to doghanchya madhye, tisrya jaagi rahava lagto.

Ek machine crore lokanna serve karu shakat nahi. Ek CPU, ek second madhe arab kadam (Part 1). Vaatun ghya: crore lok ale tar pratyekala kiti? Kaam vaatava lagta, khup machines madhe.

Machines padtat. Bijli jate, disk jalte. Sagla ekach jaagi thevla ani ti jaaga padli, tar sagla gela. Prati (copies) veglya jaagi lagtat.

Mhanun jag ashya do bhaagat vibhagla gela: tumchya hatatli machine (magnaari) ani company chya building madhli machine (denari). Ani ata sagLYAT motha prashna: ya machines ekmekashi BOLNAR kashya?

Don machines madhe taar odha, current cha pattern pathva (Part 1, Chapter 1.2: signal mhanje he ch), bits gele. Don machines zalya. Pan crore machines? Pratyekala pratyekashi taar jodli tar? Ganit kara: 100 machines la 4,950 taara lagtil, crore machines

la... jamnar nahi. (He “n squared” aahe: sankhya vadhli ki jod tya sankhyechya varga ne vadhtat. He naav lakshat theva, ya part madhe he rakshasa parat parat yeil, ani pratyek veli koni tari tyala harvun shrimant hoil.)

Upay tumhi rozchya jagat baghitla aahe: rasta. Pratyek ghara pasun pratyek ghara paryant rasta nasto; galli aste, chowk asto, highway asto. Tasach machines cha rasta banto: tumchi machine javlchya chowkala jodleli, chowk mothya chowkala, ase jod-jodun sagla jag. Ha rasta ch NETWORK aahe.

NAAV

Magnaari machine: **client**. Denari machine: **server** (serve karnari, mhanun). Jodlelya machines cha jaal: **network**. Ani “pratyekala pratyekashi jodne jamnar nahi” hi problem: **n squared problem**.

KHARYA JAGATLA EK EXAMPLE

Tumhi Zomato var order karta. Tumcha phone (client) Zomato chya server la magni pathavto. Server tumcha order thevto, restaurant chya screen la (dusra client) dakhavto, delivery walyachya phone la (tisra client) pathavto. Teen client, ek server, ani sagle ekmekanna kadhich thet jodlele nahit: pratyek jan fakt server shi bolto. Chowk ek, raste tin. N squared la asech harvtat.

ITHE LOK KAY CHUKICHE SAMAJTAT

”Majha message thet mitrachya phone la jato.” Kadhich nahi. Pratyek message aadhi company chya server la jato, mag mitrala. (End-to-end encryption asel tar company VACHU shakat nahi, pan rasta tyanchyach madhun jato: Chapter 5.6 madhe he ughdel.) Mhanunach “fukat” apps chya company kade evdhi takat aste: sagla traffic tyanchya chowkatun jato.

MAP VAR

Rupayacha rasta: tumhi mahina bharla (Level 1 pasun paisa ghusla) -> telecom company (Jio/Airtel) la, ti rasta dete. App company chya server cha kharcha ti cloud la dete (Part 5). Ani he baghaa: jya business kade SERVER aahe, tyachya kade DATA aahe; jya kade data, tyachya kade takat. Client banavne sopa ani swasta zala aahe; server chi baju hich khari malmatta aahe. Tumhi jo business banval, tyat “server var kay rahnaar” ha prashna mhanje “malmatta kay asnaar” ha prashna aahe.

SWATAHA BAGHA (5 minute)

Phone airplane mode var taka. Ata apps ughda: WhatsApp chats distat (juna data, phone var sathavlela), pan nave message yet nahit. Google Maps cha naksha adhvut ughdto, search chalat nahi. Camera purna chalto. Ata tumhala pratyek app che don bhaag distil: jo tumchya hatat aahe, ani jo koni-tari-dusrya- chya machine var aahe. Hi rekha baghayla shikne mhanje network samajne.

VICHAR KARA

1. (derivation) 100 lokanchi society aahe. Pratyekane pratyekashi thet bolayche tharvle tar 4,950 jod lagtat. Ek chowk (server) banavla tar fakt 100. Pan ata ek navin kimmat ali, ti kuthli? (Chapter 0.1 chi aathvan kara: register thevnarya var kay prashna hota?)

Uttar: Bharosa. Sagli bolni eka chowkatun jaate, mhanje chowk cha malak sagla baghto, thambvu shakto, badlu shakto. N squared sutla, pan satta ek jaagi ekavtali. Mhanunach he pattern parat parat disel: sopa rasta = madhla malak. Ani tyachya virudh che prayog pan (blockchain vagaire) yach dukhavar ubhe aahet. Tantradnyan cha pratyek design ek sauda asto: ithe suvidha vs satta ha sauda aahe. Jo he baghto, to batmya nahi, DHANDA vachto.

Chapter 5.2 [SPINE]: Pratyek machine la ek address

Network banla, raste zale. Ata post office cha juna prashna ubha rahto: patra pohochvayche tar POTTA (address) lagto. “Ram la dya” ne kaam hot nahi; Ram lakho aahe. Machines chya jagat pan tech: message pathvaycha tar tya machine cha nemka address lagto.

Address kasa asava? Part 1 ne uttar aadhich dila aahe: machine la fakt numbers samajtat. Mhanun machine cha address ek NUMBER aahe. Sadharan roop: char numbers, pratyek 0 te 255 madhla (olakhicha vaatla? 8 bits = 256 sthiti, Chapter 1.4):

142.250.183.14	(he Google chya eka server cha address)
192.168.1.5	(he tumchya ghartlya phone cha asu shakta)

Ata ek ganit kara. Char numbers, pratyek 256 prakarcha: ekun kiti address? $256 \times 256 \times 256 \times 256 =$ sadharan 430 crore. 1980 madhe he “kadhich sampnar nahit” vaatle. Aaj jagat arabo phones, TVs, gadya, ghadyala... address SAMPLE. Ha kharokhar ghadlela prasang aahe: internet chya designers ni jag kevdha hoil he kami andajle. (Upay pan aala: navin lamb address, IPv6, jya madhe address sampuch shakat nahit. Pan juna system ajun sagLikde aahe.)

Sampat aalele address jag ni kase vaparle? Ithun ek sundar yukti nighali, jee tumchya gharat aahe: tumchya WiFi router la EKACH public address milto (building la ek post-box). Ghartlya sagLya devices na router aatlya-aat aaple chhote private address deto (khli number 1, 2, 3...). Baharun sagle ekach distat; aat router hishob thevto ki kuthla jawab kuthlya device cha aahe. Ek address, dahaa machines. (Building cha watchman jasa: patra building chya naavavar yeto, watchman flat baghun pohochvto.)

NAAV

Machine cha number-address: **IP address** (Internet Protocol cha address). Juna char-number roop: **IPv4**, nava lamb roop: **IPv6**. Ghartla vaatnara-watchman: **router**. Router chya aatle address: **private IP**, jagala disnara: **public IP**.

KHARYA JAGATLA EK EXAMPLE

Cyber crime chya batmya madhe “police ni IP address varun aaropi shodhla” he vakya asta. Ata tumhala te vachta yeta: pratyek internet connection chya magey ek public IP asto, ani telecom company kade register asto ki tya veles to IP konala dila hota (Chapter 0.1 cha ledger, parat!). Mhanunach VPN naavachi cheez vikli jate: ti tumcha khara IP lapavte, dusrya deshatla dakhavte. Suraksha ani lapan-chhapan, donhi ek address chya bhovti firtat.

ITHE LOK KAY CHUKICHE SAMAJTAT

“IP address mhanje majhi kayamchi olakh.” Nahi. Tumcha public IP telecom company ROJ badlu shakte (bhadyane dilela number aahe, malkicha nahi). Ani ek IP magey shambhar lok asu shaktat (router!). Mhanun IP fakt “tya veles, tya jaagecha” pattaa aahe, aadhaar card nahi. Donhi disha ni chuk hote: lok ghabartaat “majha IP disla, sagla sampla” (nahi), ani lok nishchint pan astat “IP badalte mhanje mi lapla” (nahi: register aahe na).

MAP VAR

Kon kamavta: address SWATAHA ek malmatta zali. IPv4 address sample, mhanun juna saatha asnaryanni te VIKAYLA kadhle: ek IPv4 address aaj hazaro rupayala jato, companies chya kade lakho address che block aahet, tyanchi kimmat arabo madhe. 1990 madhe fukat vaatlele numbers aaj sona zale. Dhanda shikvan: jya cheez chi ginti THARLELI aahe ani magni vadhnaar aahe, ti cheez lavkar dharun theva. (Domains madhe he parat yeil, Chapter 5.5.)

SWATAHA BAGHA (5 minute)

Phone chya Settings madhe WiFi ughda, jodlelya network var tap kara: tithe “IP address” disel, sadharan 192.168.x.x asa. Ha tumcha PRIVATE address. Ata browser madhe search kara: “what is my IP”. Vegla number disel! Ha tumchya router cha PUBLIC address. Don address, don duniya: aat ani baher. Watchman-system tumhi swataha baghitli.

VICHAR KARA

1. (derivation) Postman la “Mumbai, Andheri, XYZ building, flat 402” asa address TUKDYA-TUKDYANI kaam karto: aadhi shahar, mag bhaag, mag building. IP address pan asach tukdyani vachla jato (aadhi motha bhaag, mag aatla). Sagle routers jagatlya SAGLYA machines chi yaadi ka thevat nahit, ha design ka nivadla?

Uttar: Karan yaadi arabo entries chi hoil ani ROJ badlel; pratyek chowkat ti thevne ani taji rakhne ashakya. Tukdya-tukdyancha design mhanje: pratyek router la fakt evdhach mahit lagta ki “ha bhaag tya dishela.” Mumbai cha postman flat 402 janat nahi, fakt Andheri kuthe te janto. Kaam vaatle gele, koni-ekala sagla janave lagat nahi. He design-tattva (hierarchy) tantradnyanat saglikde aahe: DNS madhe (5.5), company chya sanghatnet, ani AI chya aat pan. Jithe scale motha, tithe yaadi nahi, SHIDI aste.

Chapter 5.3 [DEPTH]: Message che tukde karne

(DEPTH chapter. Vagalla tari goshta pudhe jaeel, pan ha lahaan aahe ani yachya nantar “packet” ani “network slow” che khare arth ughadtat.)

Address zala. Ata message pathvaycha aahe: samja 3 MB cha photo (Part 1: mhanje sadharan 2.5 crore bits). Sopa vaatnara rasta: sagle bits ek-saath, ek lambach-lamb pravah mhanun pathva.

Ha design VAIT aahe. Ka, te tumhi truck chya udaharana ne kadhu shakta. Samja Mumbai-Pune rasta aahe ani tya varun ek 3 km lamba truck jato aahe:

1. Jya veles to jato, DUSRA konich rasta vapru shakat nahi. Ek motha message = baki saglyanchi thambleli line.
2. Truck madhe kuthe bighad zala tar? SAGLA parat pathva. 3 MB madhla shevatcha bit chukla, tar purna 3 MB parat.
3. Ani rasta ekach asel tar to padla ki sagla thambla.

Mag design tumhi kara. Rasta-yantrana kashi asavi?

Uttar: **message che LAHAAN tukde kara**. Pratyek tukdyavar liha: konala jaycha (address), konakadun aala, ani tukda number kiti (1 of 2000, 2 of 2000...). Ata pratyek tukda SWATANTRA pravaas karto. Faayde aapoaap yetat:

Raste vaatle jatat: tumchya tukdyanchya madhun dusryanche tukde pan jaatat, koni rasta adkvat nahi. Tukde VEGVEGLYA rastyani pan jau shaktat: ek highway ne, ek gaava-madhun; jo mokala tya tya kshani. Ek tukda haravla tar FAKT TOCH parat magvaycha, purna message nahi. Ani pohochlyavar number pramane lavun message parat jodla jato. (Order ulti-sulti pohochli tari chalte: numbers aahet na.)

Ekach goshta lakshat theva: tumhala vaatta tumhi “connection” ughdla aahe, ek paip, ek dhaga. Prat-yakshat DHAGA NAHI. Fakt tukde aahet, swatantra udyaa maarnare. “Connection” ha ek bhaas aahe jo donhi tokanchi software tumhala dakhavte: tukde

mojun, harvlele parat magvun, order lavun. Bhaas sundar aahe, pan to bhaas AAHE he samajlyavar network chya sagLya vichitra goshti sopya hotat: video call madhla robotic aawaz, adkleli file, “reconnecting...” sagla.

NAAV

Message cha tukda: **packet**. Tukde-karun-pathavnyachi hi yantrana: **packet switching** (1960s cha shodh, ani internet chi khari payabharni). “Harvlela parat magva, order lava, bharosa dya” he kaam karnare niyam: **TCP**. “Parat magvaycha nahi, jo pohochla to pohochla, VEGA mahatvacha” he niyam: **UDP**. (Donhi naave batmyan-madhe yetat; ata tumhala arth mahit aahe.)

KHARYA JAGATLA EK EXAMPLE

File download hotana network 2 second gela: download THAMBT0, mag tithunach pudhe chalto. Karan file la TCP: pratyek tukda pohochlach pahije, kimmat mhanun vel. Video call madhe network 2 second gela: aawaz robotic hoto, chehra atakto, pan call CHALU rahto. Karan call la UDP-sarkhe niyam: juna tukda parat magvun kay upyog? To kshan tar gela. Don design, don dhanda- garja: barobar-pana vs vega. Kuthla kadhi nivdaycha he samajne mhanje network engineer chi ardhi vidya.

ITHE LOK KAY CHUKICHE SAMAJTAT

”Net slow aahe mhanje speed kami aahe.” Adhura. Don vegLya goshti astat: ek second madhe KITI tukde jaatat (bandwidth) ani EKA tukdyala jaun-yeun kiti vel lagto (latency, Chapter 2.3 madhe bhetleli). Movie baghayla bandwidth lagte; game ani video call la latency maarte. Mhanunach “100 Mbps” cha plan asun pan game madhe “lag” yeto: tukde khup jaatat, pan pratyek tukdyacha pheraa lamba aahe. Dukandar tumhala fakt ek number vikto; khela don numbers cha aahe.

MAP VAR

Company case: packet switching ni junya telephone jagala harvle. Telephone company kade “ek call = ek rakhiv line” asa mahag design hota; packets ni tich taar hazaro lokan-madhe vaatli. Jo rasta swasta karto to jinkto: he pattern parat. Ani aaj packets chya

rastyavarchi yantre (routers) banavnari Cisco ek kaali jagatli sagLyat moulyavaan company hoti. Rasta jithe navin banto, tithe yantre viknara pahila shrimant hoto. (AI madhe aaj Nvidia tech kaam karte aahe: navin rasta, yantre viknara.)

SWATAHA BAGHA (5 minute)

Pudhchya video call chya veles network kharab zala ki LAKSH dya: aawaz robotic ka hoto? Karan software harvlele packets bharun kadhnyacha prayatna karte aahe. Chehra ka “atakto” ani mag udi maarto? Karan madhle frames (packets) gele, software ni te sodle ani navin var udi maarli. Tumhi ata bighad NAHI, DESIGN baghat aahat: vega sathi barobar- pana sodlela.

VICHAR KARA

1. (derivation) Live cricket match streaming madhe ek packet haravla. To parat magvava ka? TCP ka UDP-vichar? Ani bank transaction madhe? Donhi uttare ka veg-vegli aahet?

Uttar: Cricket: parat magvu naka. To kshan gela; parat magvun jo tukda yeil to juna asel, ani tya sathi thambne mhanje purna stream ushira. Prekshakala 2 second juna perfect picture nako, AATTACHA thoda kharab chalel. Bank: parat magva, ani jo paryant pakka pohochat nahi to paryant kahihi pudhe jau deu naka. Ek rupaya chukla tari chalnar nahi; vel lagla tari chalel. Niyam asa nighto: jithe KSHAN mahatvacha tithe vega jinkto, jithe BAROBAR-PANA mahatvacha tithe kimmat mhanun vel dyava lagto. Tumchya product madhla pratyek feature ya do rangaat vaatun baghaa; tantra-nivad aapoaap sopa hoto.

Chapter 5.4 [SPINE]: Niyam var aadhich sahmati

Address aahe, packets aahe. Pan ajun ek prashna lapla aahe, ani to sagLyat khol aahe.

Phone call chi suruvat aathva: “Hello?” “Haan, bola.” “Mi Ramesh boltoy...” He shabd mahiti det nahit; he KRAM aahe. Kon aadhi bolnaar, kasa kalel ki samor koni aahe, “aika ka” cha ishara kay. Doghanni ha kram AADHICH manlela asto, mhanun call chalte. Ek jan Japani kram ni bolla ani dusra Marathi kram ni, tar shabd pohochtil pan bolne honar nahi.

Machines madhe ha prashna ajun tikat asto, karan machine kade “samjun ghene” nahi (Part 2, Chapter 3.1: exact lagta). Don machines na bolayche tar SAGLA aadhich tharlela lagta:

- Kon suruvat karnaar? Kasa mhannaar “mi tayar aahe”?
- Message cha format kay? Kuthle bits address, kuthle maal?
- “Pohochla” kasa kalvnaar? Nahi pohochla tar kay?
- Chuk zali tar kon parat pathavnaar, kiti vela?

Ashya aadhich-tharlelya niyamanchya yaadila protocol mhantat. Ani internet mhanje kay, yacha ARDHA uttar he aahe: **internet ek vastu nahi, niyamanchi yaadi aahe**. Jo koni te niyam paalto, tyachi machine jaalat samil hote. Parvangi konachi ghyaychi nahi, fee konala dyaychi nahi. Niyam public aahe, darvaja sagLyanna ughada.

Ani niyam EKA thara var nahit, SHIDI var aahe (abstraction, parat!): sagLyat khali “taar var current kasa” che niyam, tyavar “address ani packets” che (IP), tyavar “harvlela parat magva” che (TCP), ani sagLyat var “app la kay have” che niyam. Website magvaychya niyamanche naav tumhi roj baghta: **HTTP**. Browser chya pattyat “https://...” mhanje “hi baat HTTP chya niyamani honar” evdhach. Pratyek thar fakt aaple kaam janto, khalcha thar kasa chalto he tyala mahit nasta. (Chapter 0.4: har level, khalchya cha fakt handle dharto.)

Ata dhandyachi nazar laava. Jithe protocol KHULA asto, tithe koni-hi dukan ughdu shakto: email che niyam khule, mhanun hazaro email companies. Jithe niyam EKA company che astat, tithe tichi hukumat: WhatsApp che niyam Meta chech aahet, mhanun WhatsApp la jodnara dusra app banuch shakat nahi. Khule niyam = spardha = grahakala swasta. Band niyam = kila. Doghanchehi dhande chaltat, pan konta khel khelto aahot he kalayla have.

NAAV

Aadhich-tharlele niyam: **protocol**. Web cha protocol: **HTTP** (surakshit roop: HTTPS, Chapter 5.6). Email cha: SMTP. Niyamanchi shidi: **protocol stack** ya **layers**.

KHARYA JAGATLA EK EXAMPLE

UPI he protocol aahe, app nahi. NPCI ni niyam banavle: paisa magnyacha format kay, banks ni kase bolayche, “zala” kasa kalvaycha. Niyam KHULE thevle: mhanun GPay, PhonePe, Paytm, koni-hi bank, sagle ekmekanshi chaltat. Tumhi GPay varun PhonePe walyala paise pathavta ani vichar pan karat nahi, he kevdha motha chatkar aahe: don spardhak companies chya apps madhe paisa binadhok vahto, karan protocol samaan aahe. Ani parinam jag baghta aahe: mahina 10 arab+ vyavhar. Niyam banavnara (NPCI) swataha motha “app” nahi, pan sagLya khela cha aadhaar to aahe.

ITHE LOK KAY CHUKICHE SAMAJTAT

“Internet chi malak kuthli tari company aahe (Google? Microsoft?).” Nahi. Internet konachich malmatta nahi, karan to niyamancha samuh aahe, ani niyam sagLyankade aahet. Google chya SEVA (search, YouTube) tyachya aahet; RASTA tyacha nahi. Ha farak mahatvacha aahe: seva band hou shakte, rasta band karayla jagbharche niyam badlave lagtil. (Kon niyam banavto, ha prashna Chapter 5.8 madhe.)

MAP VAR

N squared cha rakshasa parat aala hota, baghitla ka? UPI chya aadhi pratyek app la pratyek bank shi VEGLA jod banvava lagla asta: 50 apps x 300 banks = 15,000 jod. Protocol ni te 350 kele (pratyekane fakt niyamanshi jodayche). **Jo n-squared la**

protocol ni maarto, to purna bajaracha payabhut thar banto. NPCI ni he payment madhe kele. Tumchya 4-level naksha var: jithe pratyekala pratyekashi jodave lagta aahe ani sagle metaakle aahet, tithe protocol chi jaaga rikaami aahe. Ashi jaaga disli tar neat baghaa: ti arabonchi aahe.

SWATAHA BAGHA (5 minute)

Browser madhe kuthlihi website ughda ani pattyakade LAKSH dya: <https://> ne suruvat. Ata ek juni goshta: kahi varshan- purvi fakt <http://> aste (s nahi). Ata browser [http](http://) varchya site la “Not secure” mhanto. Ek akshar (s) mhanje ek purna protocol-badal, ani to badal jagbhar ka ghadla he Chapter 5.6 sangel. Aaj fakt te aksharaa dise ka baghaa.

VICHAR KARA

1. (derivation) UPI sarkha khula protocol banavnyat NPCI la (ani banks na) kay milte? Tyanni band, aaplya-malkicha system ka nahi banavla, jasa Visa/Mastercard ni banavla hota?

Uttar: Band system ni pratyek vyavhaarat fee milte (Visa cha model: dukandara kadun 1-2%). Khula system ni VAAPAR sphotasarkha vadhto: UPI fukat thevla mhanun chahawalya pasun bhajiwalya paryant sagle aale, ani 10 varshat rok-paisa (cash) cha desh digital zala. NPCI sarkari-sahakari aahe, tila fee peksha VYAAPTI havi hoti; ani banks na sagLya deshacha paisa system madhe aala he milala. Dhanda-shikvan: fee-per-vyavhaar ha ek khel aahe, purna bajarachi payabharni dusra. Doosra khel motha asto, pan to khelayla lambi saans lagte, ani kamai thet nahi tar var-chya tharatun yete (data, seva, darvaja). Google ni Android madhe hech kele hote, aathvta? Pattern ekach aahe.

Chapter 5.5 [SPINE]: Numbers aivaji naav

Address zale, niyam zale. Pan ek adchan rahili: server cha address 142.250.183.14 aahe. Tumhi “google.com” lihita. He jamte kasa? Ani mulat, ASE ka?

Karan sopa aahe: mansana numbers lakshat rahat nahit, naave rahtat (tumhi mitranche phone numbers path kelet ka? Naave save keli). Pan machines na numbers lagtat. Mhanje madhe ek PHONEBOOK lagto: naav dya, number ghya.

Ata design tumhi kara. Jagbharchya crore website sathi phonebook kasa banval?

Pahila vichar: ek motha register, eka jaagi. Chapter 0.1 chi ghanta vaajli ka? Tech prashna: (1) to register thevnarya var sagLyancha bharosa lagel, (2) jag-bharche prashna eka jaagi aale tar ti jaaga marel, (3) ti jaaga padli tar SAGLA internet “harvel” (raste chalu, pan pattec nahi).

Mhanun design vegla nivdla gela, ani to tumhi post-address madhe aadhich baghitla (Chapter 5.2): SHIDI. Naav ulta vaacha: google.com mhanje “com chya mothya vahit, google chi nond.” Pratyek tharacha register vegla: sagLyat varche register fakt he sangte ki “.com” chi yaadi kon sambhaalto; te register sangte “google.com” chi mahiti kuthe; ani tithun number milto. Koni-ekakade sagla nahi; pratyekakade aapla thar. (Hierarchy: jithe scale, tithe yaadi nahi, shidi.)

Ani vega sathi ek yukti: uttar milala ki tumcha phone ani javlche servers te LAKSHAT thevtat (thoda vel). Parat parat shidi chadhavi lagat nahi. Ya lakshat-thevnyala cache mhantat; Part 5 madhe ha shabd motha honar aahe.

Ata sagLyat sundar bhaag: naav ani number VEGLE aslya mule ek swatantrya milte. Website chi machine badla, shahar badla, cloud badla: NUMBER badlel, pan naav tech rahte. Grahakala kalat pan nahi. Dukan chi imarat badlali, pati tich rahili. Mhanunach naav (domain) hi dhandyachi malmatta aahe ani machine bhadyachi vastu.

NAAV

Naav-to-number phonebook yantrana: **DNS** (Domain Name System). Website cha naav: **domain**. Lakshat thevne: **cache**. Domain vikat ghene: registrar kade (GoDaddy vagaire), varshala thodi fee.

KHARYA JAGATLA EK EXAMPLE

2016 madhe DNS chi seva denari ek moti company (Dyn) var halla zala: lakho hacked devices ni tila prashnani budavle. Parinam: Twitter, Netflix, Spotify “band” zale, USA chya ardhya bhagasathi. Servers chalu hote, raste chalu hote, pan PHONEBOOK ughdat navhta, mhanun koni kunala shodhu shakat navhte. Tya diwshi jagala kalale ki internet chi sagLyat boring disnari yantrana kevdhi mahatvachi aahe. Payabhut goshti ashach astat: distat nahit, padlya ki sagla dista.

ITHE LOK KAY CHUKICHE SAMAJTAT

“Website mhanje domain.” Nahi. Domain fakt PATI aahe; dukan (server, code, data) vegLikde aahe. Mhanun: domain vikat ghetla mhanje website zali nahi (pati aali, dukan nahi). Ani website “band padli” tar karan pati asu shakto (domain chi fee bharli nahi: varshakathi hazaro dhande he visarun band padtat!) kinva dukan asu shakto (server padla). Don vegLyta goshti, don vegle bill, don vegle prashna.

MAP VAR

Domain ha itihasa-tla sagLyat swasta satta-kabja tharla: 1990s madhe kahi doordarshi lokanni sadhi naave (business.com, cars.com) kavdimol gheun thevli; pudhe tyach naavanche crore-crore rupaye zale (business.com sadharan 345 crore rupayala vikla gela). Aaj pan khel chalu aahe: changli naave lok aadhich gheun bastat (squatting). Chapter 5.2 chi shikvan parat: THARLELI ginti + vadhti magni = dharun theva. Ani tumchya sathi ek vyavaharik ol: dhandyachi kalpana pakki nase! tari tyacha domain 800 rupayat aadhi gheun theva; kalpana pakki hoiparyant naav gelela asta.

SWATAHA BAGHA (5 minute)

Aaplya phone chi contact-yaadi ughda. Hich tumchi DNS aahe: naave var, numbers aat. Mitrane SIM badlala tari tumhi naavach vaparta, number tumhala mahitach nasto. Ata kalpana kara contact-yaadi udali: sagle numbers chalu aahet, pan tumhi konalach phone lavu shakat nahi. DNS padne mhanje HECH, jag-bharasathi. Naave hi aathvan aahe; numbers fakt raste.

VICHAR KARA

1. (derivation) Domain varshala fakt 800-1500 rupaye ka? Evdhi mahatvachi malmatta evdhi swasta ka? (Vichar kara: kimmat kashane tharte, ani ithe scarcity nemki kashat aahe?)

Uttar: Karan NONDNI swasta aahe: register madhe ek ol lihine, jya cha kharcha lagbhag shunya (Part 1: numbers lihine fukat). Scarcity olit nahi, NAAVAT aahe: cars.com ek-ch asu shakta. Mhanun bajar don tharancha zala: nondni fee saglyanna sarkhi (swasta), pan CHANGLE naav jya kade aahe to tyachi vaatel ti kimmat magto: hi scarcity chi khari jaaga. Shikvan parat tich (Chapter 0.2): kimmat mehnati var nahi, durmilte var ubhi aste. Ani durmilita kadhi kadhi fakt “pahile pohochne” evdhich aste.

Chapter 5.6 [SPINE]: Public rastyavar private bolne

Ata ek prashna jo tumchya khishala thet bhidto. Chapter 5.3 ni sangitla: tumcha message tukdyani, DUSRYANCHYA yantran- madhun pravaas karto: telecom che routers, madhle carriers, kon-kon. Ani tumhi tya rastyane kay pathavta? Bank cha password. Card cha number. Khaajgi photo.

Postcard aathva: postman vachu shakto, karan lifafa nahi. Tumche packets pan ASECH aahet, jar kahi kele nahi tar: rasta- madhla pratyek jan vachu shakto. Mag upay? Lifafa lagto. Pan kagdi lifafa nahi; ganitacha.

Pahila vichar sopa aahe: ek gupt KILLI thareva (samja: pratyek akshar 3 ne pudhe sarkva). Mi kulup laavto, tu ughad. Madhla manus fakt kachra baghto. He juna tantra aahe (Caesar pasun World War paryant), ani he KAAM karta... ek bhayanak adchan sodun: **killi tya paryant pohochvaychi kashi?** Rasta tar public aahe! Killi rastyane pathavli tar madhla manus tich uchlel. Killi sathi bhetavech lagel... pan internet var tumhi Amazon la kadhi bhetla aahat ka? Ha prashna 1970s paryant ASODHVANIYA manla jaycha.

Mag ek kalpana aali jichya var aaj sagla online jag ubha aahe. Ashi kulup-yantrana banvli geli jyat killya DON astat, jodine:

Ek killi KULUP LAAVTE (public killi). Hi tumhi jagala vaatta, orduni sangta: “majhi kulup-killi hi ghya!”

Dusri killi UGHADTE (private killi). Hi FAKT tumchya kade. Kadhich, konalach jaat nahi.

Ata jaadu baghaa: Amazon aapli public killi sagLyanna deto. Tumhi tumcha card number tya killine kulup-band karta ani pathavta. Rasta-madhle sagle baghtat: kachra. UGHDU fakt Amazon shakto, karan ughadnari killi fakt tyachya kade. **Killi pathvaychi garajach sampli.** Kulup-killi ughad-killi pasun ganitane banvta yete, pan ulta rasta (kulup-killi varun ughad-killi kadhne) itka avghad aahe ki jagatlya sagLy machines milun pan hazaro varshat jamnar nahi (Part 1 chi aathvan: mothe numbers; ithe te dhaal banle aahet).

Browser madhla https ani kulup-icon mhanje HECH: tumchya ani tya site chya madhe hi yantrana chalu aahe. Ani WhatsApp che “end-to-end encrypted” mhanje ek paaul pudhe: killya fakt tumchya ani mitrachya phone var; MADHLI COMPANY pan vachu shakat nahi. Rasta tyancha, lifafa tumcha.

NAAV

Lifafa-ganitala **encryption** mhantat. Don-killi yantrana: **public key encryption**. Web chi surakshit vahtuk: **HTTPS**. Fakt-don-tokan-javal-killya: **end-to-end encryption**. Ani “hi site kharich ti aahe” he pramanit karnara dakhla: **certificate** (kulup-icon chya magcha kagad).

KHARYA JAGATLA EK EXAMPLE

UPI chya pratyek vyavharat hi yantrana kaam karte aahe: tumcha PIN kadhich ughda pravaas karat nahi. Ani he fakt suraksha nahi, ARTHIK PAYA aahe: lok online card/PIN takayla tayar zale KARAN he ganit tyanchya bajune ubhe aahe. Encryption nasta tar e-commerce nasta, UPI nasta, online banking nasta. Bharat chi digital economy ya ek ganiti kalpanevar ubhi aahe jichi patent kadhi konich ghetli nahi.

ITHE LOK KAY CHUKICHE SAMAJTAT

”Kulup-icon dista mhanje site bharoshachi.” NAHI. Kulup evdhach sangte ki RASTA surakshit aahe: madhla koni vachu shakat nahi. Pan palikadcha manus chor asu shakto! Scammer pan https site banavto (fee: 0 rupaye). Kulup mhanje “boli khaajgi aahe”; “samorcha khara aahe” he vegla prashna, ani to tumhalach soDvaycha asto: naav neat vaacha (amaz0n?), ghai karnarya var shanka ghya. Rasta surakshit, pravaasi tapasaa.

MAP VAR

Kon kamavta: certificates denara ek chhota udyog aahe, pan khara paisa ALIKDE aahe: encryption mule jo BHAROSA banla, tyavar arabo-kharabon cha e-commerce ubha rahila. He pattern neat baghaa, tumchya naksha sathi mahatvacha aahe: **Level 3 chi ek boring, na-disnari yantrana (encryption) Level 2 chya purna nav-ya bajarala**

(online vikri) janm dete. Payabhut thar kamavto tyā-peksha, to VAR kay ubha rahu deto tyat mothī kamai aste. Al madhe hech shodha: Al kay navin bharosa / navin bajar ubha karu deto?

SWATAHA BAGHA (5 minute)

Browser madhe bank chi site ughda, pattyat kulup-icon var click kara: “Connection is secure” ani certificate chi mahiti disel: kunala dila, kadhi sampto. Ata ek http (s nahi) site ughdaycha prayatna kara: browser lal ishara deil “Not secure”. Tumhi ata to ishara VACHU shakta: “ya rastya- varche packets ughde aahet, postcard sarkhe.”

VICHAR KARA

1. (derivation) Sarkarna kadhi kadhi vaatta: “encryption madhe amhala ek gupt magcha darvaja (backdoor) dya, fakt gunhegar pakadnyasathi.” Tantradnyanachya bajune yat kay adchan aahe? (Vichar kara: darvaja “fakt changlya lokan- sathi” asu shakto ka?)

Uttar: Ganitala hetu kalat nahi. Backdoor mhanje yantranet ek KAMJORI muddam thevne; ti kamjori police la pan ughdte ani choralā pan, ek-da sapdli ki. Ani “fakt sarkar javal killi” mhanje jagatli sagLyat mothī chori-layak vastu ek jaagi: ti jaaga halli honarach. Mhanun tantradnya mhantat: darvaja ardha ughda asu shakat nahi; ughda kinva band. Ha vaad (privacy vs surveillance) jagbhar chalu aahe ani tumhala to ata DONHI bajuni samajto: sarkarchi garaj khari aahe, pan ganiti kimmat pan khari aahe. Ashya jaagi “sopa uttar sangnara” manus ekhadi baju vikto aahe, he olkha.

Chapter 5.7 [SPINE]: Internet mhanje nakki kay

Ata sagle tukde jodta yetat. “Internet” ha shabd roj hazar vela vaparta; aaj tyala DOLYANNI baghu.

Suruvat tumchya gharatun. Tumcha phone WiFi ne router la jodlela (Chapter 5.2). Router telecom company chya (Jio/ Airtel chya) jaalala: tumchya bhagatlya khambya-varchya kinva jaminitleya fibre taarela. Ti taar shaharachya mothya kendrala. Tithun deshbharchya mothya rastyanna. He zala ek company cha JAAL.

Pan Jio cha grahak Airtel chya grahakala message pathavto, tevha? Doghanchi jaale VEGLI aahet. Uttar: companies ni aapli jaale thravik jaagi EKMEKANNA jodli aahet (mothya shahara-madhli “exchange” kendre), ani tithe saudebaji aahe: kadhi “tu majha traffic ne, mi tujha” (barabari), kadhi lahaan jaal mothyala bhade dete. **Internet mhanje HEC: hazaro veg-veglya malkanchi jaale, ekmekanna jodleli, samaan niyamani (protocols, Chapter 5.4) bolnari.** Naav pan tech sangta: inter-net, jaalan-madhla-jod. Malak konich nahi; sahbhagi sagle.

Ani deshabaher? Ithe ek dhakka basel: samudrakhali. Jagatla 99% aantar-deshiy data SAMUDRATLYA KEBALI-madhun jato: bota- evdhya jaad kachechya taara, lakho km, samudra-talashi pasarlelya. (Satellite? 1% peksha kami: mahag ani dur, mhanje latency, Chapter 2.3.) Bharatat ya kebal kinaryavar yetat: Mumbai ani Chennai hi mothi “landing” thikane. Ek kebal katli (jahajacha nangar, bhukamp) ki purna deshacha internet mand hoto: he ghadlela aahe, parat ghadel.

Ekada he disla ki “cloud” ani “WiFi” chi hawa-havi bhasha udte: internet mhanje kach, taamba, khambe, samudri kebal, buildings madhle routers. VASTU aahe, jaadu nahi. Ani vastu aahe mhanun tichi malki aahe, kimmat aahe, rajkaran aahe: kebal kon taakto, exchange kuthe aahe, he sagle satta- prashna aahet.

NAAV

Internet cha rasta viknari company: **ISP** (Internet Service Provider: Jio, Airtel, BSNL). Samudri taar: **undersea cable** (ya submarine cable). Jaale jodnyachi saudebaji: **peering**. Tumchya ghara-paryantcha shevatcha tukda: **last mile** (ha sagLyat mahag tukda asto: pratyek gharala vegli taar!).

KHARYA JAGATLA EK EXAMPLE

Jio chi goshta parat ghya, ata purna nazrene. 2016 chya aadhi Bharatat data mahag hota karan jaal LAHAAN hote. Jio ni kay kele: 2.5 lakh crore laavun NAVE jaal banavle: fibre, towers, samudri kebalinche hisse. Mag data lagbhag fukat kela (Chapter 3.7: darvaja-chaal). Parinaam: desh online aala, ani Jio darvajacha malak. Pan mul-baat hi: tyanni SEVA nahi, RASTA banavla. Rasta banavnyala vel ani arabo lagtata, mhanunach to kabja itka tikto: spardhakala parat sagle khamb ubhe karave lagtil.

ITHE LOK KAY CHUKICHE SAMAJTAT

"Internet satellite ni chalte" (nahi: 99% samudri keballi). "Cloud mhanje hawa" (nahi: cloud mhanje dusryachya buildings madhle computers, Part 5 madhe purna). "WiFi mhanje internet" (nahi: WiFi fakt tumchya ghara-atla shevatcha 10 meter cha bin-taari tukda aahe; pudhe sagla taar aahe). Teenhi chuka ekach mulatun: internet DISAT nahi, mhanun lok tyala hawa samajtat. Ata tumhala to dista: kach ani taamba.

MAP VAR

Rupayacha rasta, purna chain: tumche mahine 300 (Level 1 pasun ghusla) -> Jio (ISP) -> Jio keballi-companies na ani yantra-viknaryanna (Nokia, Ericsson, Cisco) -> te chip-walyanna (TSMC!). Ani ulat disha pan aahe: Google/Meta ATA SWATAHA samudri keballi taaktaat (2Africa, Blue-Raman: Bharatala jodnarya), karan bhade dene mahag padte ani rasta aaplya hatat asava vaatta. Jevha SEVA-company RASTA vikat ghetete, tevha samajaa: rasta hich khari satta hoti. (AI madhe aaj hech: model-companies chips ani data-centers vikat ghetayat. Pattern olkhaa.)

SWATAHA BAGHA (5 minute)

Laptop var ughda: submarinecablemap.com (fukat aahe). Mumbai var zoom kara: dahaa-pandhra kebaali kinaryala yetana distil, naavansahit. Pratyek rekha mhanje samudra-talashi padleli khari kach aahe, ani tumcha kalcha WhatsApp message kadachit tyach rekhene Amerika la gela hota. Ekda ha naksha baghitla ki “internet” shabd kadhich purvisarkha vatnar nahi.

VICHAR KARA

1. (derivation) Ek desh aaplya deshapurte internet BAND karu shakto (he ghadle aahe). Pan purna jagacha internet koni-hi band karu shakat nahi. Design madhli kuthli goshta he don-hi sangte?

Uttar: Desha-atla band shakya aahe karan desha-baher jaanare raste MOJKE astat: kinaryavarchi landing thikane ani mothe ISP; sarkar ni tya mojak-ya gala-ya aavaLlya ki desh vegla padto. Pan jagacha band ashakya, karan KENDRA NAHIC: hazaro jaale, hazaro malak, koni-ek switch nahi. He apghaataane nahi, DESIGN ne aahe: internet chi payabharni ashya yantranesathi zali hoti ji ek bhaag padla tari chalel (packets navya rastyani jaatat, Chapter 5.3). Shikvan tumchya naksha sathi: vikendrit (decentralized) yantrana maarayla avghad aste, pan tichyavar hukumat pan avghad aste. Kendrit yantrana kamvayla sopi, maarayla pan sopi. Pratyek platform ha sauda karto; tumhi pan karal.

Chapter 5.8 [DEPTH]: Internet che niyam kon banavto

(DEPTH chapter. Ha shevatcha aahe ani lahaan aahe. Yachya nantar “standard” ha shabd batmyan-madhe disla ki tumhala tyachya magcha satta-khel disel.)

Chapter 5.4 ni sangitla: internet mhanje niyam. Chapter 5.7 ni sangitla: malak konich nahi. Mag ek vichitra prashna urla: **niyam aahet, pan raja nahi. He chalta kasa? Niyam BANAUTO kon?**

Uttar ajab aahe: kahi boring sansthaa, jagbharche engineers, ani khup saarya meetings. Mukhya teen naave:

IETF: internet che mul niyam (TCP, HTTP vagaire) ithe ghadat. Ani yachi paddhat sundar aahe: KONI-HI ek prastav (proposal) lihu shakto, tyala RFC mhantat. Charcha hote, DON goshtinni prastav jinkto: sahmati ani CHALNARA code. Sadasyata nahi, fee nahi, deshacha veto nahi. 1969 pasun he ase chalta aahe.

ICANN: naave ani numbers chi vaatni (domains, IP blocks) sambhaalnari sanstha. .com kon chalvel, nave TLDs (.shop, .ai) kadhi yetil, he ithe tharte.

W3C ani browser-companies: web che niyam (HTML vagaire). Ani ithe ek khara sach: kaagdavar niyam samiti banavte, pan VYAVHARAT niyam te hotat je mothe browsers (Chrome!) khare laagu kartat. Google chya browser kade 60%+ vaata aahe, mhanje Google chya nirnayanna niyamanchi takat aapoaap yete.

Mhanje chitra ase: niyam khule aahet, prakriya lokshahi- sarkhi aahe, PAN takat barabar vaatleli nahi: jya company kade vaapar-kartya (users) jaast, tichya aawajala vajan jaast. Mhanun mothya companies ya boring meetings madhe aaple pagaari engineers pathavtat: standard chya table var basne mhanje bhavishyacha bajar aakhne.

He lamb vaatla tar ek Bharatiy udaharan sagla sangel: UPI che niyam NPCI banavte: ani mhanunach Bharat ne “aapla protocol aapan banavne” (DPI: UPI, Aadhaar-stack, ONDC) ha rashtriya khel kela aahe. Niyam banavnara bajar aakhto; niyam gheNARA bajarat fakt khelto.

NAAV

Sarva-manyas niyam: **standard**. Prastavachi paddhat: **RFC**. Sanstha: **IETF** (mul niyam), **ICANN** (naave/numbers), **W3C** (web). Bharatacha sarkari-protocol khel: **DPI** (Digital Public Infrastructure).

KHARYA JAGATLA EK EXAMPLE

SMS aathva: to pan ek standard hota, telecom-walyanni banav- lela. WhatsApp aala ani SMS la khaun taakla: karan WhatsApp cha “standard” ek-companycha hota, vegane badalta yet hota, ani fukat hota. Mag Google ni SMS cha varasdaar (RCS) pudhe dhakalla, Apple ne varsha-n-varshe to NAKARLA (iMessage cha kila!), ani 2024 madhe akher ghetla. Ya purna ladhait grahak kuthech navhta; ladhai standards chi hoti, karan jo standard jinkto tyachi seva “default” hote, ani default hone mhanje arabo grahak fukat milne.

ITHE LOK KAY CHUKICHE SAMAJTAT

”Kuthli tari jagtik sanstha (UN sarkhi) internet chalavte.” Nahi. UN chi ek sanstha (ITU) juni telecom sambhaalte, pan internet che khare niyam varchya khulya sansthaan-madhe ghadtat, ani kahi desh (jyanna kendrit niyantran have) he badlaycha prayatna satat kartat: “internet ITU khali aana” he tyanche magne asta. Mhanje “kon niyam banavto” ha prashna samplela nahi; ti ek CHALU ladhai aahe, ani tichyat Bharat ek motha aawaj aahe (arabo users + swataha che DPI).

MAP VAR

Tumchya 4-level naksha var hi shevatchi ol: **Level 3 chya khaali ajun ek maala aahe: NIYAM. Jo niyam banavto, to Level 3 chya sagLya khelanchi zameen aakhto.** Ani niyam banavnyat paisa THET nahi (IETF fukat, NPCI na-nafa), pan niyamanchya var ubhe rahnare bajar arabonche astat, ani niyam-banavnaryachya deshala/companyla tyas bajarat ghar-cha maidaan milte. Mhanun: batmyan-madhe “standard”, “protocol”, “regulation” he shabd disle ki thamba ani vaacha. Kantalvane vaattat, pan tithe pudhchya dashakachi malki tharte aahe. Al madhe HICH ladhai aatta chalu aahe: Al che niyam kon banavnaar? (Book 2 madhe bhétel.)

SWATAHA BAGHA (5 minute)

Google madhe search kara: “RFC 1149”. Ughdel to ek KHARA, adhikrut RFC aahe: kabutaran-varun internet packets pathav- nyacha (vinodi) prastav, 1 April la lihilela. Ani lok ni to 2001 madhe KHARA karun pan dakhavla (9 packets, 1 kabutar). He baghnyacha uddesh: RFC kiti khule aahe he janavne: vinod pan chaltoy, ani tya-ach paddhatine jagacha sagla internet pan ghadla aahe. Khuli prakriya ashi diste.

VICHAR KARA

1. (derivation) Bharatane UPI banavla ani to Singapore, UAE, France ashya deshan- madhe pan neto aahe. “Protocol niryaat karne” (export) madhe Bharatala kay milte? Paisa tar thet milat nahi. (Chapter 5.4 chi NPCI-shikvan pudhe nya.)

Uttar: Zameen milte. Jya deshat UPI-dhartiche niyam laagle, tithe Bharatiy banks, apps, ani kaushalya (jya lokanna he chaltva yeta) ghar-chya maidanavar kheltat. Ani jagtik vyavhaarat vajan milte: aaj paise-pathavnyache jagtik niyam Visa/Mastercard/SWIFT (paschimi) aakhtat; pratyek desh jo UPI gheto, to tya ekadhikarala thoda kamjor karto ani niyam-tablavar Bharatachi khurchi mothi karto. He Level 3 che bhu-rajkarana (geopolitics) aahe: 20vya shatakat desh TEL ani bandaran-sathi ladhle; 21vya madhe kebaali, chips ani PROTOCOLS sathi ladhtat. Tumhi jo naksha banavla aahe (4 levels), to deshanchya pattlivar pan tantotant laagto: hich tyachi shevatchi pariksha.

PART 3 CHA SHEVAT: NAKASHA ANI PUDHCHA PAAUL

Je tumchya-kade ata aahe

CLIENT / SERVER	magnari machine / denari machine; malmatta server chya bajula aste
N SQUARED	pratyekala pratyekashi jodne ashakya; jo he chowk/protocol ni sodavto to malak hoto
IP ADDRESS	machine cha number-pattaa; public vs private (router = building cha watchman)
PACKETS	message che numbered tukde, swatantra pravaas; "connection" ha fakt bhaas aahe
TCP vs UDP PROTOCOL	barobar-pana vs vega: file vs video call aadhich tharlele niyam; internet mhanje vastu nahi, niyamancha samuh (UPI = protocol!)
DNS	naav -> number cha phonebook; shidi + cache; domain = pati, server = dukan
ENCRYPTION	public rastyavar ganiti lifafa; don killya (public kulup laavte, private ughadte); kulup = rasta surakshit, manus nahi!
INTERNET	hazaro malkanchi jaale, jodleli; 99% samudri keballi; kendra nahi, malak nahi
NIYAM	IETF/ICANN/W3C + mothya companies; standard banavnara bajar aakhto (Bharatacha DPI)

Ek pariksha, swataha sathi

Pustak na ughadta, bolun uttar dya:

1. WhatsApp message thet mitracha phone la ka jaat nahi?
2. Video call madhe aawaz robotic ka hoto, pan file download madhe file kadhich adhuri ka yet nahi?
3. Kulup-icon (https) kay SANGTO ani kay SANGAT NAHI?
4. UPI la "protocol" ka mhanave, app ka nahi? Ani tyane Bharatala kay dile?

Adkal tar: 1 -> 5.1, 2 -> 5.3, 3 -> 5.6, 4 -> 5.4/5.8.

Part 4 madhe kay aahe

Server la crore lokancha data milto. To THEVAYCHA kuthe? Ek crore madhun EK nond shodhaychi kashi, dolyachya pheratit? Don lok EKAACH veles ek goshta badaltat tevha kay hote? Ani sagla data udala tar? Part 4: DATA AANI AATHAVNI: database che jag, jithe pratyek mothya company chi khari sampatti padleli aste.

PART 3 CHA MINI-GLOSSARY

cache	uttar lakshat thevne, parat shidi na chadhnyasathi (5.5)
certificate	"hi site kharich ti aahe" cha dakhla (5.6)
client / server	magnari / denari machine (5.1)
DNS	naav-to-number phonebook, shidi-rachnecha (5.5)
domain	website che naav; bhadyachi nond, malmatta pati (5.5)
DPI	Bharatacha sarkari-protocol khel: UPI, Aadhaar (5.8)
encryption	ganiti lifafa; vachu fakt killi-wala shakto (5.6)
end-to-end	killiya fakt don tokenvar; madhli company pan andhali (5.6)
HTTP / HTTPS	web cha protocol / tyache surakshit roop (5.4, 5.6)
ICANN	naave ani numbers vaatnari sanstha (5.8)
IETF / RFC	mul niyam banavnari sanstha / khula prastav (5.8)
IP address	machine cha number-pattaa (IPv4 june, IPv6 nave) (5.2)
ISP	internet-rasta viknari company: Jio, Airtel (5.7)
last mile	tumchya ghara-paryantcha shevatcha (mahag) tukda (5.7)
latency	ek pheraa kiti vel; bandwidth peksha vegli goshta (5.3)
n squared	jod sankhyechya vargane vadhtat hi problem (5.1)
network	jodlelya machines cha jaal (5.1)
packet	message cha numbered, swatantra tukda (5.3)
peering	don jaalanchi "tu majha, mi tujha" saudebaji (5.7)
private key	ughadnari killi; kadhich konala nahi (5.6)
protocol	aadhich tharlele bolnyanche niyam (5.4)
public key	kulup laavnari killi; jagala vaatleli (5.6)
router	packets pudhe dhakalnara chowk; gharcha watchman (5.2)
standard	sarva-many a niyam; banavnara bajar aakhto (5.8)
TCP / UDP	barobar-pana-wale / vega-wale niyam (5.3)
undersea cable	samudri kach-taar; jagacha 99% data ithun (5.7)